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International shipments of *Wolbachia*-infected mosquito eggs: towards the scaling-up of World Mosquito Program operations

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Summary

The *Wolbachia* insect control method, employed by the World Mosquito Program (WMP), relies on introgressing *Wolbachia* through target *Aedes aegypti* populations to reduce the incidence of dengue. Since 2010, the WMP has been producing *Wolbachia*-infected mosquitoes at numerous sites across the globe for release in 11 countries. As the technology has matured, greater focus has been placed on mosquito production at larger central facilities for transport to remote release sites, both domestically and internationally.

Of particular note is the production of *Wolbachia*-infected mosquitoes at the WMP's Australian production facility for successful international deployments in Fiji, Vanuatu, Kiribati and Sri Lanka. This requires careful management of both production and supply-chain processes to ensure that the quality of the mosquito eggs, specifically the hatch rate and *Wolbachia* infection rate, is maintained.

To ensure the cost-effectiveness and scalability of the *Wolbachia* method, these processes will be further refined to facilitate deployment from large centralised production facilities.

Keywords

Aedes aegypti – Australia – Dengue – Egg shipment – Mosquito production – *w*Mel – *Wolbachia* – World Mosquito Program.

Introduction

The *Wolbachia* method involves the introgression of *Wolbachia pipientis* through target *Aedes aegypti* populations to reduce the incidence of vector-borne diseases caused by flaviviruses and alphaviruses, such as dengue, Zika, chikungunya and yellow fever [1, 2]. This method relies on the production, transport, and release of *Wolbachia*-infected *Ae. aegypti,* which facilitates introgression through a target *Ae. aegypti* population.

Wolbachia pipientis, an obligate endosymbiotic alphaproteobacterium, has been shown to be dispersed globally nearly everywhere and is estimated to infect up to 66% of insect species [3, 4, 5]. The *Wolbachia* strain *w*Mel, derived from *Drosophila melanogaster*, was stably introduced into the yellow fever mosquito, *Ae. aegypti*, in 2011 [2]. The *w*Mel strain confers three signature traits on *Ae. aegypti* that enable *Wolbachia*-infected *Ae. aegypti* to be deployed as a public health intervention [2]. The first two, *Wolbachia*-mediated cytoplasmic incompatibility and maternal transmission, facilitate the population introgression of *w*Mel into a target *Ae. aegypti* population. The third signature trait of *w*Mel, when present in *Ae. aegypti,* is that it interferes with viral replication. *Wolbachia*-infected *Ae. aegypti* have significantly lower vector competence (ability to transmit a pathogen) for all four serotypes of dengue virus, as well as for chikungunya, Zika, yellow fever and Mayaro virus, when compared to mosquitoes not infected with *Wolbachia* [6, 7, 8, 9, 10]. Moreover, *w*Mel is genomically and phenotypically stable in field deployments [11, 12, 13].

The World Mosquito Program (WMP) (formally 'Eliminate Dengue') first demonstrated the feasibility of the *Wolbachia* method as a public health intervention in 2011, through the introgression of the *w*Mel *Wolbachia* strain into naive *Ae. aegypti* populations, using large semi-field cages located at James Cook University, Queensland, Australia [2]. Subsequently, *Wolbachia* (*w*Mel) has been fixed in *Ae. aegypti* populations throughout northern Queensland, Australia, via releases of *Wolbachia*-infected mosquitoes [14, 15]. The

WMP *Wolbachia* method is being deployed in 11 countries, with release sites in Mexico and South America (Colombia and Brazil) and South-East Asia (Sri Lanka, Indonesia and Vietnam), as well as the Oceania region (Australia, Kiribati, Fiji, Vanuatu and New Caledonia) (Figure 1). Moreover, a non-WMP consortium has established *Wolbachia* in *Ae. aegypti* populations within Malaysia [16].

The *Wolbachia* method has been shown to be highly successful. In all areas where *Wolbachia* has been established, and public health data are available, it has been demonstrated that the *w*Mel *Wolbachia* strain reduces the incidence of dengue [14, 15, 17, 18, 19]. This included a randomised controlled trial in Yogyakarta, Indonesia, that demonstrated a 77% reduction in dengue incidence and an 86% reduction in dengue hospitalisation in clusters that had been randomly selected to receive *w*Mel deployments, when compared to untreated clusters [19].

The WMP approach in all regions is heavily dependent on community support and participation. Typically, community engagement will precede any planned release and releases will not begin until a community acceptance threshold is reached [17]. The WMP also involves local communities through educational programmes in schools, information events and community-facilitated releases, in which local communities help, directly or indirectly, with the deployment.

As with all aspects of WMP's *Wolbachia*-infected *Ae. aegypti* release, the methods for gaining community acceptance and support have evolved over time as the scale of the releases has increased. Initially, releases placed a much higher emphasis on approaches to individual households. As release sites grow in size and number, and with the approval of local regulatory agencies, the Public Acceptance Model (PAM) will be employed.

The Public Acceptance Model looks to gain acceptance at the community level, rather than at the individual level [20]. At present, PAM has only been employed in a limited fashion but, as deployments become larger, PAM will become the dominant strategy. Regardless of the method employed, the rights of participating communities to be informed, raise concerns, and get involved in WMP projects are always respected. Releases of *Wolbachia*-infected *Ae. aegypti* only ever occur with sufficient local awareness, understanding and support.

Wolbachia-infected mosquito deployment by the WMP relies on one of two methods for mosquito release:

- adult mosquitoes housed in release tubes or cups
- mosquito eggs deployed in mosquito-release containers (MRCs).

The choice of method depends heavily on the local climate and geography, as well as the socio-economic environment, with some release sites using mixed strategies. However, both methodologies require the availability of high-quality, *Wolbachia*-infected *Ae. aegypti* eggs. Mass production of *w*Mel-infected *Ae. aegypti* eggs for the WMP is taking place in rearing facilities in Medellin, Colombia; Rio de Janeiro, Brazil; and Melbourne, Australia. Production is also

Figure 1

Countries in which *Wolbachia*-infected *Aedes aegypti* have been released by the World Mosquito Program (WMP)

As of 2022, the WMP and local partners have released *Wolbachia*-infected *Aedes aegypti* eggs in 11 countries

occurring in several smaller, localised production facilities in New Caledonia, Mexico, Vietnam and Indonesia.

As the number and remoteness of *Wolbachia* release locations increase, deployments increasingly rely on the eggs being packaged at centralised production facilities and subsequently transported to the target sites. Transportation of *Wolbachia*-infected *Ae. aegypti* eggs requires rapid, temperature-controlled shipping, and must also comply with biosecurity containment requirements. The WMP has successfully used a strategy that relies on production in the Monash facility in Melbourne, Australia. The eggs are then transported to release sites across the globe. This review will focus on the methodology used for transport from this production facility to Fiji, Vanuatu, Kiribati and Sri Lanka.

Production and storage

The manufacturing process for *Wolbachia*-infected *Ae. aegypti* requires strict quality assurance to ensure that mosquitoes for release are fit for purpose, comply with all local guidelines, and pose no risk due to inadequate safety or quality. Throughout the manufacturing process, regardless of location, the WMP employs a quality management system that is continually being refined. Reflecting this commitment, the WMP Monash facility has recently received ISO 9001:2015 certification [21]. These physical and procedural processes are documented to ensure strict compliance. Moreover, all staff members are trained and regularly assessed to ensure adherence to documented practices.

World Mosquito Program production is conceptually divided into aquatic rearing, adult rearing, egg processing and material supply (i.e. eggs and adult mosquitoes) (Figure 2). In addition, these processes are replicated over three levels of rearing numbers:

- a brood level with low-density rearing conditions for stock maintenance
- an amplification level with intermediate mosquito density to produce large numbers of eggs to feed the next level
- a mass-production level, capable of producing the quantity of eggs needed for release.

There are quality assurance checks throughout this process to ensure consistency of materials (typically eggs and/or adult insects), and continuous quality control to ensure effective facility operations. These checks include *Wolbachia* frequency during larval rearing, blood pathogen screening before feeding, and a colony fertility test at the end of production.

Although *Ae. aegypti* eggs can survive for extended periods of time, *Wolbachia*-infected *Ae. aegypti* eggs have a comparatively reduced lifespan [22, 23, 24, 25]. This poses a major hurdle for the centralised production and transport of eggs for mosquito release. Furthermore, storage conditions can have a major impact on the fitness of offspring derived from *Wolbachia*-infected eggs [26].

For these reasons, the WMP has laid down a set of storage and transport conditions that allow *Wolbachia*-infected eggs to be stored for up to eight weeks post-production, as well as developing a logistical chain for transporting eggs to release sites in Kiribati, Fiji, Vanuatu and Sri Lanka.

When *w*Mel-infected eggs were stored between 18–22°C, with a relative humidity of 70–80%, viability losses (hatch-rate decline) of only 20% were observed over an eight-week period. Furthermore, *Wolbachia* density in offspring was not affected for the duration of storage [27]. To facilitate transport, the WMP contracted a specialist courier company that could provide stable shipping conditions (temperature and humidity), as well as logistical support to manage import and custom requirements. With this service, eggs could be delivered weekly to release sites. Fitness tests (egg hatch rate, *Wolbachia* density in offspring and wing length of offspring), conducted concurrently at Monash and the destinations mentioned above, demonstrated no adverse fitness effects because of shipping or storage.

Egg shipment to multiple locations

Regulatory pathway and export process mechanism

The manufacturing process involves moving *Wolbachia*-infected mosquitoes across the globe. To ensure containment of *Wolbachia*-infected *Ae. aegypti*, eggs are shipped either in capsules or on paper to the target countries. Rapid transit from a production facility to the release site is achieved by working with local and international biosecurity agencies and transport companies. *Wolbachia*-infected *Ae. aegypti* present regulatory advantages over some other technologies. They have been repeatedly defined as not being genetically modified (GM) and, as such, not subject to international or domestic GM regulatory frameworks [28, 29, 30].

However, *Wolbachia*-infected mosquitoes are still subject to biosecurity regulatory requirements and must be handled correctly. The shipment process is highly specific, as every country operates its own biosecurity regulatory framework. Typically, each country requires both an export and import permit for *Wolbachia*-infected *Ae. aegypti* transport. These permits are guided by a pre-established authorisation from the relevant biosecurity authorities. Depending on the specifics of individual shipments, transit permits may also be required.

Although pre-existing agreements may be in place, each shipment typically requires its own permits and certification.

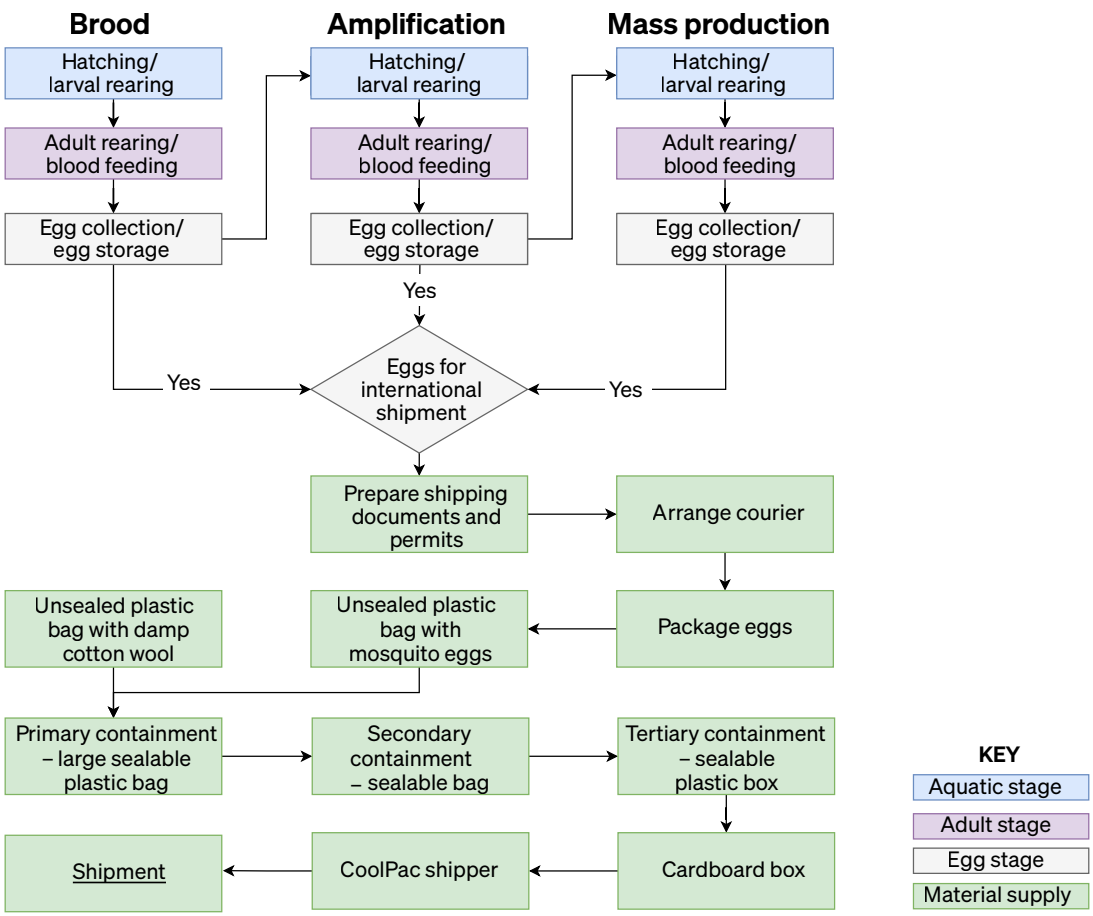

Figure 2

Flow chart of the production and shipment of *w*Mel *Wolbachia*-infected *Aedes aegypti* by the World Mosquito Program

Production is divided into three levels with increasing mosquito density: brood, amplification and mass production. Although eggs from each stage can be used for deployments, typically only those from mass production are packaged for transport. Shipment requires that *Wolbachia*-infected *Aedes aegypti* eggs are contained in several layers to ensure compliance with biosecurity and maintain quality

To facilitate rapid transport and minimise risk, the WMP undertakes partnerships with logistics and support companies.

Within Australia, WMP partners with LabCabs (Murarrie, Queensland, Australia) to facilitate shipments from the Melbourne-based production facility to field sites. The WMP typically employs CoolPac shippers (Doveton, Victoria, Australia) for all egg shipments. The WMP packaging ensures that *Wolbachia*-infected *Ae. aegypti* eggs are contained within six layers (Figure 3). Each batch of eggs is given an internal identification number, which is linked to its *Wolbachia* screening, its human blood pathogen report and its equine blood pathogen report.

This comprehensive supply-chain management ensures that the cause of any adverse event, from early development to release, can be rapidly identified and corrected. This product traceability, together with careful monitoring, ensures product integrity throughout all WMP operations.

As there is no generic regulatory pathway to provide oversight of transport, the WMP handles this process on a release-site-by-release-site basis. Typically, this involves engaging with regulatory agencies in both the country of manufacture (in this case, Australia) and the country importing the material. The process relies on three separate types of approval – permits for import, export, and transit. Each of these permits ensures that biosecurity is maintained.

Import permits are required for shipping mosquitoes from the proposed release site to a production facility for strain development, backcrossing, and the transfer of *Wolbachia*-infected release material to sites in the importing country.

Export permits are required to ensure appropriate containment procedures are followed. However, they are also vital in many countries, such as Brazil, where strong protections exist to prevent the illegal export of biological samples.

Transit permits are required to ensure that the appropriate procedures are followed when transporting mosquitoes. For example, as there are no practical direct flights from Australia to Kiribati, and the eggs typically pass through Fiji,

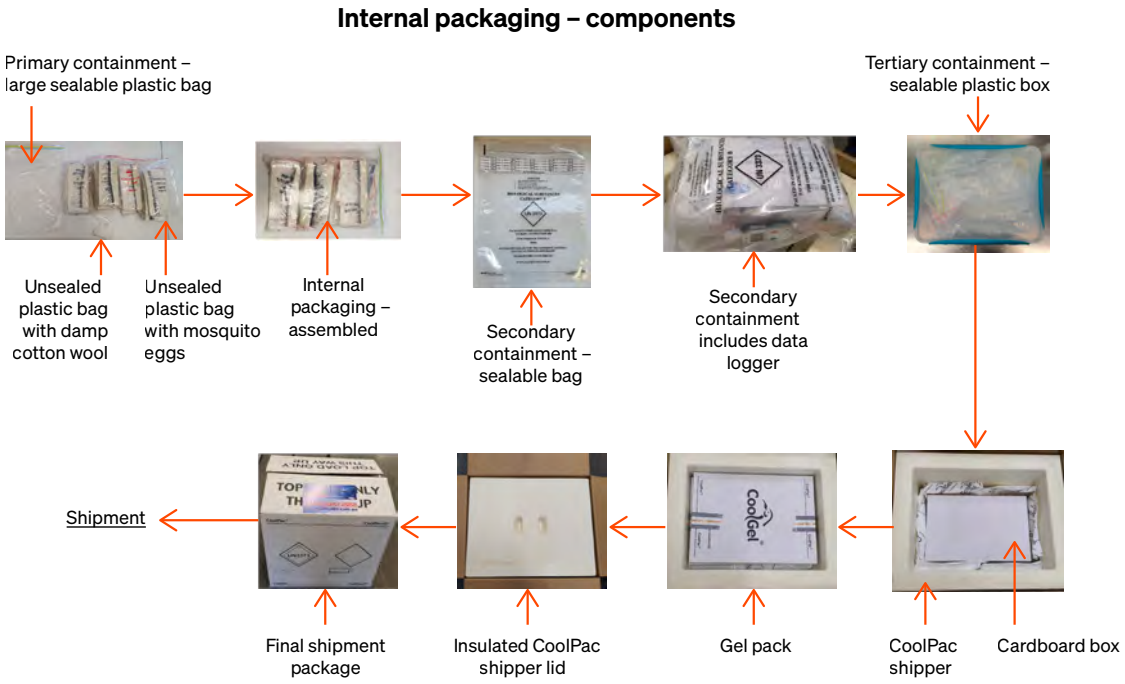

Figure 3

Flow chart of shipment packaging and containment for mosquito eggs shipped from the World Mosquito Program's Melbourne production facility

The eggs are packaged to maintain optimal viability through transit. The internal packaging contains a packet of damp cotton wool that provides the eggs with a humid environment to minimise desiccation. The CoolPac Polyurethane Insulated Shipper box is a multi-use cold chain packaging system that uses an insulated container and refrigerant (gel packs) to ensure that the contents are maintained at 15–25°C. The Australian biosecurity regulatory framework requires that all shipped material is appropriately contained to prevent the risk of quarantine material escaping. The World Mosquito Program ships its mosquito eggs within six levels of containment, including a solid plastic box, to prevent the accidental release of material

Wolbachia-infected *Ae. aegypti* for release need a transit permit from the Fijian authorities for their transport.

The WMP almost exclusively transports eggs, rather than adults, of both *Wolbachia*-infected and uninfected *Ae. aegypti*. Transporting eggs provides a strong barrier to escape. Additionally, in this species, the eggs are more durable than the adult stages in terms of exposure to adverse conditions during shipment. Moreover, since they are less bulky and not as heavy as adults, their shipment costs are lower.

Projects: Monash to Sri Lanka, Vanuatu, Fiji and Kiribati

The Melbourne-based WMP production facility has shipped adapted strains of eggs to field sites across the western Pacific (Fiji, Kiribati and Vanuatu) and South-East Asia (Sri Lanka). From 2018 until 2021, the WMP regularly shipped eggs to each of these countries. The LabCab (Australia) shipments ensured rapid transportation to each of these release sites, typically taking only 24–48 hours to Sri Lanka but up to five days to Kiribati, due to its more remote location. The LabCab service includes door-to-door shipment. Moreover, the company also provides considerable assistance with navigating customs procedures.

Costs associated with the international transport of mosquito eggs can vary, with the common factors that affect logistics (fuel prices, package size, etc.) fluctuating regularly. The specific requirements mandated by the port of entry and/or export may also influence the cost of shipment. Regulatory requirements for export permits and potential additional certification incur a fee for each individual shipment. For example, transportation to Kiribati was, on average, twice the cost of shipping to Fiji. This was primarily due to the lack of direct shipment, requiring transportation through an intermediate country and thus involving additional regulatory/compliance hurdles.

Finally, ongoing factors associated with the COVID-19 pandemic have greatly hampered almost all transportation and release. This included disruption of the Sri Lankan releases due to the inability to safely conduct and monitor field operations, thus forcing their stoppage. Once releases continued, the transportation costs had risen by approximately 25%. It is currently unknown when, or even if, the supply chain and transportation will return to normal. However, it seems almost certain that there will be ongoing increases in the costs associated with the global transportation of materials in the foreseeable future.

Typically, reductions in cost can be achieved by decreasing the frequency of shipments. However, the decision to ship more eggs less frequently is dictated by the facility's weekly rearing capacity, the robustness of the material being shipped and the on-site storage capacity.

Conclusions

The WMP has successfully deployed *Wolbachia*-infected *Ae. aegypti* in 11 countries. These deployments have resulted in widespread reductions in the incidence of dengue. Initially, releases were of locally produced mosquitoes that involved limited transport to deployment sites. However, as the technology has matured, greater emphasis has been placed on production from central facilities, which is then transported to more remote international sites. Refinement of the process has enabled the successful transport of *Wolbachia*-infected mosquito eggs from the production facility in Monash University (Melbourne, Australia) to release sites in Fiji, Vanuatu, Kiribati and Sri Lanka.

Establishing production facilities can mean overcoming considerable financial and regulatory hurdles. Moreover, small facilities lack the efficiencies needed for cost-effective deployment of the *Wolbachia* method. For this reason, it remains highly impractical to establish facilities in every release area, especially if the lifespan of a particular project is unknown. Thus, the WMP has developed the necessary frameworks and processes for global distribution from centralised facilities, using transport to Pacific release sites as a pilot programme. The most critical element of this is stability, both in terms of production output and the quality of the product.

Inconsistency in the number of *Wolbachia*-infected eggs supplied could have serious negative effects, both in the cost and effectiveness of *Wolbachia* introgression at a target site. Therefore, throughout the deployments described above, consistent egg production was maintained through constant refinement and monitoring of the processes involved. This culminated in ISO 9001 certification of the Monash production facility. Although this is not a requirement, the authors would encourage any parties looking to undertake centralised production to investigate the feasibility of certification. ISO 9001 provides a management framework that greatly facilitates harmonious, high-quality output of the target product.

Likewise, maintaining egg quality throughout the transport chain is critical for effective deployment. Sharp declines in egg viability (hatch rate) or *Wolbachia* infection rates could also adversely affect deployment. Thus, it was crucial to carry out quality assurance/quality control checks at each stage of transportation. To facilitate consistency, transportation partners and highly specific packing were used. Moreover, a considerable body of work has been carried out to define the optimal conditions for storage and transport. Typically, no significant declines in either the hatch rates or *Wolbachia* infection rates of the eggs transported to Fiji, Vanuatu, Kiribati and Sri Lanka were observed when compared to eggs from the same batch stored at the production facility. Given production and logistical costs, deployment may continue, on a case-by-case basis, for egg batches with low hatch rates. However, diminished *Wolbachia* infection rates would probably result in the destruction of the failed batch. Moreover, any failures will also trigger reviews of the facility's processes.

The WMP will constantly review these processes to ensure that consistency is maintained as it moves towards large-scale, centralised production. A key focus of this ongoing review will be improvements in the cost effectiveness of transporting the eggs. Some of this will likely come from scaling up; however, the WMP must also achieve savings in costs per shipment if the programme's goals are to be achieved. There are several proposed options for packaging the eggs, aimed at reducing costs in either their transportation or deployment.

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Expéditions internationales d'œufs de moustiques infectés par *Wolbachia* : vers une intensification des opérations du World Mosquito Program

J.A. Denton, D.A. Joubert, A.A. Goundar & J.R.L. Gilles

Résumé

La méthode de contrôle des insectes mise en œuvre par le World Mosquito Program (WMP) recourt à la bactérie *Wolbachia* et repose sur l'introgression de cette dernière par le biais des populations cibles d'*Aedes aegypti* dans le but de réduire l'incidence de la dengue. Depuis 2010, le WMP produit des moustiques infectés par *Wolbachia* à partir de nombreux sites répartis dans le monde entier en vue de leur lâcher dans 11 pays. Cette technologie s'étant bien développée, l'accent est désormais mis sur la production de moustiques dans des établissements centralisés de plus grande envergure afin de les expédier vers des sites de lâcher éloignés, tant sur le territoire national qu'à l'échelle internationale.

Il importe de souligner la production de moustiques infectés par *Wolbachia* conduite par le WMP sur son site australien et leur déploiement international couronné de succès aux Fidji, au Vanuatu, à Kiribati et au Sri Lanka. Cela nécessite une gestion rigoureuse des procédures aussi bien lors de la production que tout au long de la chaîne d'approvisionnement afin de veiller au maintien de la qualité des œufs de moustiques et plus spécifiquement du taux d'éclosion et du taux d'infection par *Wolbachia.*

Afin d'assurer la rentabilité et l'évolutivité de la méthode basée sur *Wolbachia*, ces procédures seront encore affinées afin de faciliter le déploiement à partir de sites de production à grande échelle centralisés.

Mots-clés

Aedes aegypti – Australie – Dengue – Expédition d'œufs – Production de moustiques – *w*Mel – *Wolbachia* – World Mosquito Program.

Envíos internacionales de huevos de mosquito infectados por *Wolbachia*: rumbo a la extensión de las actividades del World Mosquito Program

J.A. Denton, D.A. Joubert, A.A. Goundar & J.R.L. Gilles

Resumen

El método de control de insectos con *Wolbachia* que emplea el World Mosquito Program (WMP) reposa en la introgresión de *Wolbachia* a través de poblaciones de *Aedes aegypti* para reducir la incidencia del dengue. Desde 2010, el WMP viene generando mosquitos infectados por *Wolbachia* en numerosas instalaciones repartidas por todo el globo, mosquitos que después son liberados en 11 países. A medida que la tecnología ha ido madurando, la tendencia ha sido cada vez más la de concentrar la producción de mosquitos en grandes instalaciones centrales y desde allí enviarlos a los remotos lugares de suelta, ya estén en el mismo país o en otros países.

Especialmente destacable es la producción de mosquitos infectados por *Wolbachia* en las instalaciones que el WMP tiene en Australia, utilizadas con éxito como centro de operaciones para aplicar el método en otros países como Fiji, Vanuatu, Kiribati y Sri Lanka. Ello exige una cuidadosa gestión de los procesos tanto de producción como de la cadena de suministro, que garantice un nivel constante de calidad de los huevos de mosquito, y más concretamente de las tasas de eclosión y de infección por *Wolbachia*.

A fin de asegurar que el método ofrezca una buena relación costo-eficacia y sea reproducible a mayor escala, esos procesos serán perfeccionados para facilitar el despliegue de los insectos a partir de grandes instalaciones de producción centralizada.

Palabras clave

Aedes aegypti – Australia – Dengue – Envío de huevos – Producción de mosquitos – *w*Mel – *Wolbachia* – World Mosquito Program.

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